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Robust trajectory planning for UAV communication systems in the presence of jammers

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Abstract Unmanned Aerial Vehicle (UAV) has emerged as a promising novel application for the Sixth-Generation (6G) wireless communication by leveraging more favorable Line-of-Sight (LoS) propagation. However, the jamming resistance by exploiting UAV's mobility is a new challenge in the UAV-ground communication. This paper investigates the trajectory planning problem in an UAV communication system, where the UAV is operated by a Ground Control Unit (GCU) to perform certain tasks in the presence of multiple jammers with imperfect power and location information. To ensure the reliability of the GCU-to-UAV link, we formulate the problem as a non-convex semi-infinite optimization, aiming to maximize the average worst-case Signal-to-Interference-plus-Noise Ratio (SINR) over a given flight duration by designing the robust trajectory of the UAV under stringent energy availability constraints. To handle this problem efficiently, we develop an iterative algorithm for the solution with the aid of *S-procedure* and Successive Convex Approximation (SCA) method. Numerous results demonstrate the efficacy of our proposed algorithm and offer some useful design insights to practical system.
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1. Introduction

With the advancement in the Sixth-Generation (6G) communication systems, there is an urgent need to push the envelope of the performance to accommodate the surging traffic demands, which poses stringent requirements on both the radio resources and existing communication infrastructures.[1,2] Equipped with intelligent sensors and precise Global Positioning System (GPS), Unmanned Aerial Vehicles (UAVs) are gaining ever-increasing popularity in the field of 6G wireless communication due to their highly controllable maneuverability,

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on-demand coverage and Line-of-sight (LoS) links.[3–7] Compared with the terrestrial communication system, deploying UAVs in the air to complete certain tasks is envisioned as a promising solution to enhance the performance of the existing communication networks. On one hand, the LoS channel facilitates the establishment of high-rate communication between the UAV and Ground Control Unit (GCU). On the other hand, the communication quality can be improved significantly since the wireless channel from the GCU to UAV is dominated by LoS with generally a small path loss. Due to these desirable features, a proliferation of interests has recently been inspired from both the academic and industrial communities.[8–11]

According to the actual role of UAVs in the communication network, prior works can be roughly divided into two parts, namely UAV-assisted communication and cellular-enabled UAV communication. In UAV-assisted communication, the UAV usually serves as a new aerial communication device to provide services for ground users. To serve edge users and alleviate data traffic from the GCU, Ref. 12 conceived a novel offloading technique to maximize the energy efficiency of UAVs by jointly optimizing communication scheduling, user partitioning, and flying trajectory. To alleviate the performance bottleneck caused by the "doubly near-far" phenomenon, the authors in Ref. 13 advocated a Radio Frequency (RF) Wireless Power Transfer (WPT) scheme in UAV network, and formulated a joint resource allocation and trajectory planning strategy to maximize the system throughput, subject to both UAVs' maximum speed and energy constraints. In Ref. 14, a UAV-assisted mobile relaying system was investigated, wherein the throughput of the ground node was improved with the optimization of transmit power along with the relay trajectory. Considering limited energy budget for propulsion and communication, Ref. 15 jointly optimized the power and 3D trajectory to minimize the outage probability for the UAV-enabled relaying system to the ground users. On the other hand, UAVs can be effectively manipulated by the GCU to perform various missions in the cellular-enabled UAV system. Specifically, Ref. 16 investigated the UAV trajectory optimization design to minimize the mission completion time while ensuring the information retrieval with a high probability, subject to the UAV's initial and final location constraints, the speed constraint, as well as a minimum received Signal-to-interference-plus-noise Ratio (SINR) constraint. Furthermore, a joint time–frequency scheduling and power allocation framework is proposed in Ref. 17 to guarantee the reliable signals reception in the uplink communication, where many UAVs are controlled by a GCU to carry out various tasks. Meanwhile, in order to eliminate the co-channel interference and maximize the UAV network throughput, Ref. 18 invoked a novel interference cancellation strategy to fully utilize the limited backhaul links among GCUs in the cellular network.

Despite the alluring benefits come from UAVs, the broadcasting nature of radio propagation brings severe challenge to the communication system. In fact, the control signal reception of UAV is not only affected by the quality of communication channels, but also vulnerable to any potential threats (e.g., eavesdropping and jamming). Hence, fully exploring the anti-jamming technology in UAV network can provide some effective ways to improve the communication performance.[19–22] In particular, Ref. 19 advocated the usage of a novel smart attack game between a UAV and subjective jammers, and adapted a reinforcement-learning-based power allocation strategy to increase the secrecy capacity of the network. In Ref. 20, the authors provided an analytical approach to optimize a UAV's trajectory to combat jamming signals and provide high-quality services to ground nodes. What's more, some practical studies for jamming resistance technology of multi-UAV formation were proposed in Refs. 21,22. To provide reliable communication and efficient control in the GCU-controlled-UAV network, Ref. 21 investigated an UAV-priority-based resource coordination method to reduce the impact of jammers and improve Quality of Service (QoS) of the system. Inspired by,[21] Ref. 22 fully analyzed the distinctive characteristics of different interferences, and subsequently devised a powerful resource allocation strategy to maximize the minimum SINR among the UAV swarm.

In addition, some existing works also assumed that jammers' information was imperfectly known (e.g., Refs. 23-25), which is more suitable for the realistic scene. Ref. 23 proposed a joint robust trajectory and power design for a single-UAV network with multiple jammers located in uncertain regions. To provide different QoS requirements for diverse applications, Ref. 24 investigated the throughput-maximum problem in the UAV-assisted communication system, where multiple ground users are scheduled to communicate with multi-UAV swarm in the presence of jammers with uncertain location information. In Ref. 25, the authors considered the worst-case secrecy rate maximum problem in an UAV-aided cognitive radio network relying on joint resource allocation and trajectory optimization.

Motivated by the above observations, this paper focuses on designing the trajectory for efficient control and reliable communication in a GCU-controlled-UAV network. Specifically, we consider a practical scenario where a GCU controls a UAV to perform certain tasks with multiple jammers distributed across the regions. It is assumed that the jammers' power and location are partially known to the UAV, subject to norm-bounded estimated errors. Accordingly, the quality of wireless link largely depends on the UAV's position (i.e., trajectory) over time. The goal of this paper is to design robust flight trajectory to reduce the impact of jammers and improve the quality of control signals. In comparison to the prior works, to the best of our knowledge, this is the first work that conceives the energy constraint along with the mobility constraint to maximize the average worst-case SINR in a UAV network. However, it is quite challenging to solve the intractable optimization problem, due to the highly non-convex and semi-infinite nature of the objective function. The key contributions are listed as follows:

(1) We present a general optimization framework for a UAV network in the presence of multiple jammers with uncertain power and location information. An average worst-case SINR maximization problem is proposed to provide reliable QoS of the received signals by robust trajectory design under stringent energy availability constraints.
(2) To tackle the formulated intractable problem, we first recast this semi-infinite problem into an equivalent but more compact form with the aid of slack variables and S-Procedure, and then develop an efficient iterative algorithm to obtain the solution based on SCA technique.

(3) We perform in-depth experiments under diverse system parameter configurations. The simulations clearly demonstrate the efficacy of our proposed method. The results also shed light on the design and performance analysis of the UAV communication system.

The reminder of this paper is organized as follows. Section 2 presents the UAV communication model and formulates the corresponding trajectory planning problem of interest. In Section 3, a low-complexity iterative algorithm is proposed for the considered problem. The simulation results are provided in Section 4. Finally, this article is concluded in Section 5.

Notations. In this paper, scalars are denoted by lower case, vectors and matrics are respectively denoted by boldface lower-case and boldface upper-case letters. $\mathbf{R}^{M\times N}$ denotes the set of M - by - N real-valued matrics, and $\mathbf{H}^{M}$ denotes the set of M - by - M Hermitian matrics. For a vector $\boldsymbol{a}$, $||\boldsymbol{a}||$ represents its Euclidean norm and $\boldsymbol{a}^{\mathrm{T}}$ denotes its transpose.

2. System model

2.1. Network architecture and communication model

An uplink wireless UAV communication system is considered in Fig. 1, where a GCU controls a UAV to carry out certain tasks with $M \geqslant 1$ jammers distributed across the region. In such scenario, the GCU periodically sends control signals to UAV at each time slot. Simultaneously, multiple jammers are sending jamming signals to interfere with legitimate communication. In this paper, we consider the locations in Three-dimensional (3D) Cartesian coordinate system with all dimensions measured in meters (m). Suppose that the GCU locates at $[x_{\mathrm{s}}, \mathrm{y}_{\mathrm{s}}, 0]$, and the location of each jammer $m \in \mathscr{M} \triangleq \{1, 2, ..., M\}$ is $[x_{\mathrm{j},m}, y_{\mathrm{j},m}, 0]$, where $\boldsymbol{w}_{\mathrm{s}} = [\mathrm{x}_{\mathrm{s}}, \mathrm{y}_{\mathrm{s}}]^{\mathrm{T}} \in \mathbf{R}^{2\times 1}$ and $\boldsymbol{w}_{\mathrm{j},m} = [x_{\mathrm{j},m}, y_{\mathrm{j},m}]^{\mathrm{T}} \in \mathbf{R}^{2\times 1}$ represent the horizontal locations of the GCU and jammer m, respectively. The objective of the UAV is to ease the interference caused by jammers and enhance the quality of control signals, by designing UAV's flight trajectory over a finite flight duration T. For this purpose, an optimization strategy is required. Before stating the problem formulation, we first present the models for jammers' uncertain information, UAV trajectory, GCU-to-UAV transmission channel, and energy consumption.

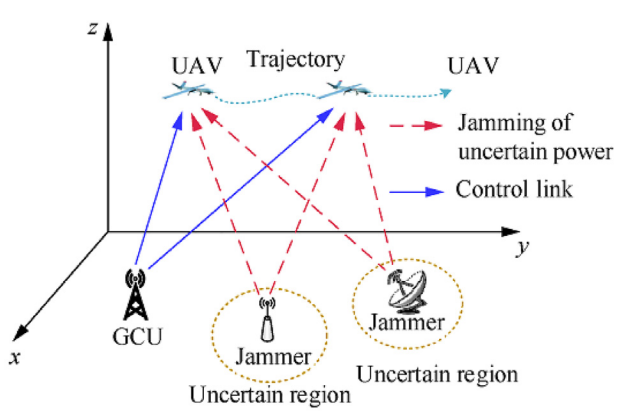

Fig. 1 Uplink UAV communication scenario.

2.1.1. Bounded error model

According to the works in Refs. 23,24, in practice, the UAV knows the exact information (e.g., location, power) of the GCU via proper information exchange, but the accurate information of jammers is uncertain due to the existence of estimation errors and quantization errors. It is assumed that the potential location region and power level can be estimated by the UAV. This assumption is reasonable from two aspects. On one hand, owing to the LoS link in communication systems, it is possible for UAV to obtain the approximate location of jammers through the camera or Synthetic Aperture Radar (SAR).[25] On the other hand, the UAV can detect carrier energy to acquire the approximate jammers' power by applying for carrier detection technology.[26] As a result, we can adopt the bounded error model to characterize the relation between the actual and estimated information (i.e., location, power) of jammers, which can be represented as

$$\boldsymbol{w}_{\mathrm{j},m} = \tilde{\boldsymbol{w}}_{\mathrm{j},m} + \Delta\boldsymbol{w}_{\mathrm{j},m},\ \forall m \tag{1a}$$

$$\mathscr{A}_{\mathrm{j},m} \triangleq \left\{\Delta\boldsymbol{w}_{\mathrm{j},m} = \left[\Delta x_{\mathrm{j},m}, \Delta y_{\mathrm{j},m}\right]^{\mathrm{T}} \in \mathbf{R}^{2\times 1} :\ \|\Delta\boldsymbol{w}_{\mathrm{j},m}\|^2 \leqslant \varepsilon_m^2\right\} \tag{1b}$$

$$p_{\mathrm{j},m} = \tilde{p}_{\mathrm{j},m} + \Delta p_{\mathrm{j},m},\ \forall m \tag{2a}$$

$$\psi_{\mathrm{j},m} \triangleq \left\{\Delta p_{\mathrm{j},m} \in \mathbf{R} :\ \|\Delta p_{\mathrm{j},m}\| \leqslant \xi_m\right\} \tag{2b}$$

where $\tilde{\boldsymbol{w}}_{\mathrm{j},m}$ and $\tilde{p}_{\mathrm{j},m}$ denote the estimates of the actual location $\boldsymbol{w}_{\mathrm{j},m}$ and power $p_{\mathrm{j},m}$, respectively; $\Delta\boldsymbol{w}_{\mathrm{j},m}$ and $\Delta p_{\mathrm{j},m}$ represent the estimation location error and power error with the bound of ε_m and ξ_m, respectively; $\mathscr{A}_{\mathrm{j},m}$ and $\psi_{\mathrm{j},m}$ express the uncertainty of regions and power, respectively.

2.1.2. UAV trajectory model

In the sequel, we elaborate the trajectory model of UAV. It is assumed that a UAV flies from a pre-determined starting point $[x_{\mathrm{I}}, \mathrm{y}_{\mathrm{I}}, H]$ to the endpoint $[x_{\mathrm{F}}, \mathrm{y}_{\mathrm{F}}, H]$ to execute certain missions over a finite time T. We define $\boldsymbol{q}_{\mathrm{I}} = [x_{\mathrm{I}}, y_{\mathrm{I}}]^{\mathrm{T}} \in \mathbf{R}^{2\times 1}$ and $\boldsymbol{q}_{\mathrm{F}} = [x_{\mathrm{F}}, y_{\mathrm{F}}]^{\mathrm{T}} \in \mathbf{R}^{2\times 1}$ as the horizontal initial and final coordinate. Note that the UAV should not ascend and descend frequently in consideration of avoiding unnecessary energy consumption, thereby we assume the flying altitude of UAV is a fixed value H, which corresponds to the minimum altitude required for obstacles avoidance.[27] For analytical simplicity, the total flight time T is equally discretized into N slots, indexed by $n \in \mathscr{N} \triangleq \{1, 2, ..., N\}$. Therefore, the slot duration is given by $\delta = T/N$, which is chosen sufficiently small such that the UAV is assumed to be stationary within each time slot.[17] As such, the horizontal trajectory of UAV can be approximately characterized as $\boldsymbol{q}[n] = [x[n], y[n]]^{\mathrm{T}} \in \mathbf{R}^{2\times 1}$. Let $\boldsymbol{u}[n] = [u_x[n], u_y[n]]^{\mathrm{T}} \in \mathbf{R}^{2\times 1}$ denote the speed vector at the UAV over the whole flight duration, and $\boldsymbol{a}[n] = [a_x[n], a_y[n]]^{\mathrm{T}} \in \mathbf{R}^{2\times 1}$ denote the corresponding acceleration vector. Considering the limited mobility of UAV, it strictly satisfies the following constraints:

$$\boldsymbol{q}[n] = \boldsymbol{q}[n-1] + \boldsymbol{u}[n-1]\delta + \frac{1}{2}\boldsymbol{a}[n-1]\delta^2,\ n = 2, 3, ..., N \tag{3a}$$

$$\boldsymbol{u}[n] = \boldsymbol{u}[n-1] + \boldsymbol{a}[n-1]\delta,\ n = 2, 3, ..., N \tag{3b}$$

$$\boldsymbol{q}[1] = \boldsymbol{q}_1 + \boldsymbol{u}_1\delta + \frac{1}{2}\boldsymbol{a}_1\delta^2 \tag{3c}$$

$$\boldsymbol{q}[N] = \boldsymbol{q}_{\mathrm{F}}, \boldsymbol{u}[N] = \boldsymbol{u}_{\mathrm{F}} \tag{3d}$$

$$\| \boldsymbol{a}[n] \| \leqslant a_{\max}, \ \forall n \tag{3e}$$

$$\| \boldsymbol{u}[n] \| \leqslant u_{\max}, \ \forall n \tag{3f}$$

$$\| \boldsymbol{u}[n] \| \geqslant u_{\min}, \ \forall n \tag{3g}$$

where $\boldsymbol{q}_{\mathrm{I}}$, $\boldsymbol{u}_{\mathrm{I}}$ and $\boldsymbol{a}_{\mathrm{I}}$ indicate the initial location, speed and acceleration respectively; $\boldsymbol{q}_{\mathrm{F}}$ and $\boldsymbol{u}_{\mathrm{F}}$ represent the final location and speed respectively; $u_{\min}$, $u_{\max}$ and $a_{\max}$ denote the minimum flying speed, maximum speed and maximum acceleration respectively.

2.1.3. Ground-to-air channel model

As recorded in practical tests,[28,29,7] the ground-to-air communication channels can be well dominated by the LoS links, especially when the UAV flies at a relatively moderate altitude from the certain environment with little blockage and scattering. In addition, it is assumed that the Doppler effect caused by UAV's mobility can be perfectly compensated based on existing techniques.[30] As a result, we adopt the widely-used large scale path-loss model for the LoS links between the GCU to UAV. Consequently, the channel gain between the GCU and UAV in time slot n is given by

$$h_{\mathrm{s}}[n] = \beta_0 d_{\mathrm{s}}^{-\alpha}[n] = \frac{\beta_0}{\left(\| \boldsymbol{q}[n] - \boldsymbol{w}_{\mathrm{s}} \|^2 + H^2\right)^{\alpha/2}} \tag{4}$$

where $d_{\mathrm{s}} = \sqrt{\| \boldsymbol{q}[n] - \boldsymbol{w}_{\mathrm{s}} \|^2 + H^2}$ is the distance between the UAV and GCU, β_0 denotes the channel power gain at the unit distance (1 m), and $\alpha \geqslant 2$ represents the path loss exponent determined by the radio propagation environment. Similarly, when the distance between the UAV and jammer m is $d_{\mathrm{j},m} = \sqrt{\| \boldsymbol{q}[n] - \boldsymbol{w}_{\mathrm{j},m} \|^2 + H^2}$, the corresponding channel power gains in time slot n can be expressed as

$$h_{\mathrm{j},m}[n] = \beta_0 d_{\mathrm{j},m}^{-\alpha}[n] = \frac{\beta_0}{(|\boldsymbol{q}[n] - \boldsymbol{w}_{\mathrm{j},m}|^2 + H^2)^{\alpha/2}} \tag{5}$$

2.1.4. Energy consumption model

In the practical UAV flight scenario, the total energy consumption is composed of two parts. The first one is the communication-related energy, which is used to maintain signaling exchange between the UAV and GCU. The other part is the propulsion energy, which is required for UAV to remain aloft and maintain its maneuverability. In fact, the communication-related energy is far less than the propulsion energy, so it is often ignored (e.g., Refs. 27,31). Based on the above assumption, the energy consumption of the UAV during N time slots is given by

$$E(\boldsymbol{q}[n]) = \delta \sum_{n=1}^{N} \left[c_1 ||\boldsymbol{u}[n]||^3 + \frac{c_2}{||\boldsymbol{u}[n]||} \left(1 + \frac{||\boldsymbol{a}[n]||^2}{g^2} \right) \right] + \Delta_k \tag{6}$$

where c_1 and c_2 are two parameters related to the aerodynamics, g indicates the gravitational acceleration constant, and $\Delta_k = 0.5J(\boldsymbol{u}_{\mathrm{F}}^2 - \boldsymbol{u}_1^1)$ denotes the change of kinetic energy of the UAV, whose value is determined by the initial speed, final speed, as well as the UAV's mass J.

2.2. Problem formulation

This section formally presents the problem formulation. Since the information of each jammer is incomplete, based on the above bounded error model, the worst-case received SINR of the UAV at time slot n can be specialized as

$$\gamma[n] = \min_{\substack{\Delta \boldsymbol{w}_{\mathrm{j},m} \in \mathcal{A}_{\mathrm{j},m}, \\ \Delta p_{\mathrm{j},m} \in \psi_{\mathrm{j},m}}} \frac{p_0 h_{\mathrm{s}}[n]}{\sum_{m=1}^{M} p_{\mathrm{j},m} h_{\mathrm{j},m}[n] + \sigma^2} \tag{7}$$

where p_0 denotes the transmit power of the GCU, $p_{\mathrm{j},m}$ denotes the statistical value of jammer m's transmitting power, and σ^2 represents the power of Additive White Gaussian Noise (AWGN) at the receiver.

Our objective is to maximize the UAV's average worst-case SINR over N time slots, by jointly optimizing its trajectory $\boldsymbol{q}[n]$, speed $\boldsymbol{u}[n]$ and acceleration $\boldsymbol{a}[n]$, subject to the UAV mobility constraints and energy consumption constraints. Mathematically, the investigated problem can be formulated as

$$(\mathrm{P1}) \max_{\{q[n],u[n],a[n]\}} \frac{1}{N} \sum_{n=1}^{N} \min_{\substack{\Delta w_{\mathrm{j},m} \in \mathcal{A}_{\mathrm{j},m}, \\ \Delta p_{\mathrm{j},m} \in \psi_{\mathrm{j},m}}} \frac{p_0 h_s[n]}{\sum_{m=1}^{M} p_{\mathrm{j},m} h_{\mathrm{j},m}[n] + \sigma^2} \tag{8a}$$

$$\text{s.t.} \quad \delta \sum_{n=1}^{N} \left[c_1 \| u[n] \|^3 + \frac{c_2}{||u[n]||} \left(1 + \frac{\| a[n] \|^2}{g^2} \right) \right] + \Delta_k \leqslant \Gamma \tag{8b}$$

$$\boldsymbol{q}[n] = \boldsymbol{q}[n-1] + \boldsymbol{u}[n-1]\delta + \frac{1}{2}\boldsymbol{a}[n-1]\delta^2, \ n = 2, 3, ..., N \tag{8c}$$

$$\boldsymbol{u}[n] = \boldsymbol{u}[n-1] + \boldsymbol{a}[n-1]\delta, \ n = 2, 3, ..., N \tag{8d}$$

$$\boldsymbol{q}[1] = \boldsymbol{q}_{\mathrm{I}} + \mathbf{u}_{\mathrm{I}}\delta + \frac{1}{2}\mathbf{a}_{\mathrm{I}}\delta^2 \tag{8e}$$

$$\boldsymbol{q}[N] = \boldsymbol{q}_{\mathrm{F}}, \boldsymbol{u}[N] = \boldsymbol{u}_{\mathrm{F}} \tag{8f}$$

$$\| \boldsymbol{a}[n] \| \leqslant a_{\max}, \ \forall n \tag{8g}$$

$$\| \boldsymbol{u}[n] \| \leqslant u_{\max}, \ \forall n \tag{8h}$$

$$\| \boldsymbol{u}[n] \| \geqslant u_{\min}, \ \forall n \tag{8i}$$

where in Eq. (8b) means the energy consumption constraint, and Γ indicates the maximum propulsion energy allowed for the UAV over the whole flight duration. Moreover, Eqs. (8c)–(8i) represent the trajectory, speed, and acceleration constraints of the UAV.

Note that problem P1 is challenging to be solved optimally for the following two aspects. First, the nonlinear objective function and the constraints Eq. (8b) and Eq. (8i) render the problem non-differentiable and non-convex. In addition, due to the uncertainty in each jammers' location, P1 is an intractable semi-infinite optimization problem, which makes the solution more difficult. In short, we are faced with an intractable non-convex semi-infinite problem, which is usually considered as

NP-hard. In what follows, we will perform a series of transformations for P1 and propose an efficient algorithm to solve it.

3. Proposed solution

In this section, we propose an efficient iterative algorithm to solve problem P1. Specially, we first simplify the non-smooth objective function into a tractable form. After that, for the semi-infinite numbers of constraints evoked by jammers' imperfect information, we provide some insights to recast the sophisticated constraints into an equivalent but more enunciable form by applying for S-Procedure algorithm. Moreover, for the resulting non-convex problem, we develop an iterative algorithm for the solution with the assistance of slack variables and SCA method. In the end, we summarize the algorithm and evaluate its computation complexity.

3.1. Reformulation of objective function in Eq. (8a)

It is readily seen that, the higher the jammers' power is, the worser the SINR of the UAV will achieve. Motivated by this, we can conduct a simple mathematical manipulation by setting the $p_{\mathrm{j},m}$ as maximum power (i.e., $\tilde{p}_{\mathrm{j},m}+\xi_m$) to eliminate the variable $\Delta p_{\mathrm{j},m}$ in the objective function Eq. (8a). Furthermore, for problem Eq. (P1), each jammer's uncertain location parameter (i.e., $\Delta \boldsymbol{w}_{\mathrm{j},m}$) only exists in the expression of each accumulation term in the denominator. By plugging Eq. (4) and Eq. (5) into Eq. (8a), problem Eq. (P1) can be transformed into the following equivalent form:

$$(\mathrm{P2})\ \max_{\{\boldsymbol{q}[n],\boldsymbol{u}[n],\boldsymbol{a}[n]\}} \frac{1}{N}\sum_{n=1}^{N} \frac{p_0 \frac{\beta_0}{(\|\boldsymbol{q}[n]-\boldsymbol{w}_{\mathrm{s}}\|^2 + \mathrm{H}^2)^{\alpha/2}}}{\sum_{m=1}^{M}(\tilde{p}_{\mathrm{j,m}}+\xi_m)\frac{\beta_0}{\min\limits_{\Delta \boldsymbol{w}_{\mathrm{j,m}}\in \mathscr{A}_{\mathrm{j,m}}}(\|\boldsymbol{q}[n]-\boldsymbol{w}_{\mathrm{j,m}}\|^2 + H^2)^{\alpha/2}}+\sigma^2} \tag{9a}$$

$$\text{s.t. Eqs.(8b)} - \text{(8i)} \tag{9b}$$

Note that, the difficulty remains to be in the sophisticated expression of the objective function Eq.(9a). To address this issue, we apply slack variables to derive an explicit expression of problem Eq. (P2) by using the following lemma.

Lemma 1. Introducing slack variables $\boldsymbol{I}\triangleq[I[1], I[2], ..., I[N]]$ and $\boldsymbol{L}\triangleq[L[1], L[2], ..., L[N]]$, an optimization problem equivalent to Eq.(P2) can be obtained as follows:

$$(\mathrm{P3})\ \max_{\{\boldsymbol{q},\boldsymbol{u},\boldsymbol{a},\boldsymbol{I},\boldsymbol{L}\}} \frac{1}{N}\sum_{n=1}^{N}\frac{p_0\beta_0}{I[n]L[n]} \tag{10a}$$

$$\text{s.t.}\left(\|\boldsymbol{q}[n]-\boldsymbol{w}_{\mathrm{s}}\|^2+H^2\right)^{\alpha/2} \leqslant I[n], \forall n \tag{10b}$$

$$\sum_{m=1}^{M}\left(\tilde{p}_{\mathrm{j},m}+\xi_j\right)\frac{\beta_0}{\min\limits_{\Delta \mathbf{w}_{\mathrm{j},m}\in A_{\mathrm{j},m}}\left(\|\boldsymbol{q}[n]-\boldsymbol{w}_{\mathrm{j},m}\|^2+H^2\right)^{\alpha/2}}+\sigma^2 \leqslant L[n], \forall n \tag{10c}$$

$$\text{Eqs. (8b)} - \text{(8i)} \tag{10d}$$

Proof. We can prove this by contradiction. Specifically, if there are $\left(\|\boldsymbol{q}[n]-\boldsymbol{w}_{\mathrm{s}}\|^2+H^2\right)^{\alpha/2} = I[n], \forall n$ *and* $\sum_{m=1}^{M}\left(\tilde{p}_{\mathrm{j},m}+\xi_{\mathrm{j}}\right)\beta_0\left[\Delta\mathbf{w}_{\mathrm{j},m}\in A_{\mathrm{j},m}^{\min}\left(\|\boldsymbol{q}[n]-\boldsymbol{w}_{\mathrm{j},m}\|^2+H^2\right)^{\alpha/2}\right]^{-1}+\sigma^2 = L[n], \forall n$, the optimal solution to the problem Eq. (P2) and Eq. (P3) are equivalent. It is assumed that the equations in Eq.(10b) and Eq.(10c) are not strictly true, i.e., there exists $n\in\mathscr{N}$ such that $\left(\|\boldsymbol{q}[n]-\boldsymbol{w}_{\mathrm{s}}\|^2+H^2\right)^{\alpha/2} \leqslant I[n]$ and $\sum_{m=1}^{M}\left(\tilde{p}_{\mathrm{j},m}+\xi_{\mathrm{j}}\right)\beta_0\left[\Delta\mathbf{w}_{\mathrm{j},m}\overset{\min}{\in} A_{\mathrm{j},m}\left(\|\boldsymbol{q}[n]-\boldsymbol{w}_{\mathrm{j},m}\|^2+H^2\right)^{\alpha/2}\right]^{-1}+\sigma^2 = L[n]$, we can always decrease $I[n]$ and $L[n]$ to improve the objective function such that the equation holds true. As a result, the problem Eq. (P2) and (P3) are equivalent.

Noting that the term $(I[n]L[n])^{-1}$ in the objective function makes the problem P3 non-convex and difficult to be solved optimally. Consequently, we can obtain a lower bound of the objective function by using the following lemma.

Lemma 2. For any given feasible point $(I_{\mathrm{fea}}[n], L_{\mathrm{fea}}[n])$, $(I[n]L[n])^{-1}$ is lower bounded by

$$\tilde{\gamma}_{\mathrm{fea}}[n] = \frac{1}{I_{\mathrm{fea}}[n]L_{\mathrm{fea}}[n]} - \frac{1}{I_{\mathrm{fea}}^2[n]L_{\mathrm{fea}}[n]}(I[n]-I_{\mathrm{fea}}[n]) - \frac{1}{L_{\mathrm{fea}}^2[n]I_{\mathrm{fea}}[n]}(L[n]-L_{\mathrm{fea}}[n]) \tag{11}$$

Proof. Since $f(x,y)=1/xy,\ (x,y>0)$ is jointly convex with respect to x and y, the first-order Taylor expansion offers a global under-estimator at a given feasible point $(x_{\mathrm{fea}}, y_{\mathrm{fea}})$, *i.e.*,

$$\frac{1}{xy} \geqslant \frac{1}{x_{\mathrm{fea}}y_{\mathrm{fea}}} - \frac{1}{x_{\mathrm{fea}}^2 y_{\mathrm{fea}}}(x-x_{\mathrm{fea}}) - \frac{1}{y_{\mathrm{fea}}^2 x_{\mathrm{fea}}}(y-y_{\mathrm{fea}}) \tag{12}$$

Thus, by setting $x=I[n]$ and $y=L[n]$, Lemma 2 is proved.

3.2. Reformulation of semi-infinite constraint Eq. (10c)

By introducing slack matrix $\boldsymbol{s}=[s_m[n]], \forall m,n$, constraint Eq. (10c) can be equivalently substituted by

$$\sum_{m=1}^{M}(\tilde{p}_{\mathrm{j},m}+\xi_{\mathrm{j}})\frac{\beta_0}{(S_m[n])^{\alpha/2}}+\sigma^2 \leqslant L[n], \forall n \tag{13}$$

$$\min_{\Delta w_{\mathrm{j},m}\in\mathscr{A}_{\mathrm{j},m},} \|\boldsymbol{q}[n]-\boldsymbol{w}_{\mathrm{j},m}\|^2+H^2 \geqslant s_m[n], \forall n,m \tag{14}$$

It can be proved by the similar theory to Lemma 1, so we omit it in this paper. Note that the difficulty remains to be in the infinite inequality constraint Eq. (14) due to the uncertainty of the jammers' location information. Specifically, owing to the continuous nature of $\mathscr{A}_{\mathrm{j},m}$, it has an infinite number of $(\Delta x_{\mathrm{j},m}, \Delta y_{\mathrm{j},m})$ in constraint Eq. (14). By plugging Eq. (1a) and Eq. (1b) into Eq. (14), we can obtain the following constraints:

$$\Delta x_{\mathrm{j},m}^2 + \Delta y_{\mathrm{j},m}^2 - \varepsilon_m^2 \leqslant 0,\ \forall m \tag{15}$$

$$-\left(x[n]-\Delta x_{\mathrm{j},m}-\tilde{x}_{\mathrm{j},m}\right)^2-\left(y[n]-\Delta y_{\mathrm{j},m}-\tilde{y}_{\mathrm{j},m}\right)^2-H^2+s_m[n] \leqslant 0,\ \forall n,\ m \tag{16}$$

To address this issue, we hereby apply the S-Procedure[32] to the constraints Eq. (15) and Eq. (16), by using the following lemma.

Lemma 3 (*(S-Procedure):*). We denote $f_i(\boldsymbol{x})=\boldsymbol{x}^{\mathrm{T}}\boldsymbol{A}_i\boldsymbol{x}+\boldsymbol{b}_i\boldsymbol{x}+c_i, i\in\{1,2\}$, where $\boldsymbol{x}\in\mathbf{R}^{M\times 1}$, $\boldsymbol{A}_i\in\boldsymbol{H}^M$, $\boldsymbol{b}_i\in\mathbf{R}^{M\times 1}$ and $c_i\in\mathbf{R}\ominus$. The relation $f_1(\boldsymbol{x})\leqslant 0\Rightarrow f_2(\boldsymbol{x})\leqslant 0$ holds if and only if there exists $\lambda\geqslant 0$ such that

$$\lambda\begin{bmatrix}\boldsymbol{A}_1 & \boldsymbol{b}_1\\ \boldsymbol{b}_1^{\mathrm{H}} & c_1\end{bmatrix}-\begin{bmatrix}\boldsymbol{A}_2 & \boldsymbol{b}_2\\ \boldsymbol{b}_2^{\mathrm{H}} & c_2\end{bmatrix}\geqslant\mathbf{0} \tag{17}$$

According to Lemma 3, the implication (15) $\Rightarrow$ (16) holds if and only if there exists $\lambda_m[n]\geqslant 0$ such that

$$\boldsymbol{\Phi}(x[n],y[n],s_m[n],\lambda_m[n])\geqslant 0,\ \forall n,\ m \tag{18}$$

where

$$\boldsymbol{\Phi}(x[n],y[n],s_m[n],\lambda_m[n])=\begin{bmatrix}\lambda_m[n]+1 & 0 & \tilde{x}_{\mathrm{j},m}-x[n]\\ 0 & \lambda_m[n]+1 & \tilde{y}_{\mathrm{j},m}-y[n]\\ \tilde{x}_{\mathrm{j},m}-x[n] & \tilde{y}_{\mathrm{j},m}-y[n] & -\varepsilon_m^2\lambda_m[n]+c_m[n]\end{bmatrix}$$

and $c_m[n]=x^2[n]-2x[n]\tilde{x}_{\mathrm{j},m}+\tilde{x}_{\mathrm{j},m}^2+\text{'}y^2[n]-2y[n]\tilde{y}_{\mathrm{j},m}+\tilde{y}_{\mathrm{j},m}^2+H^2-s_m[n]$. By introducing slack variables $\boldsymbol{\lambda}=\{\lambda_m[n],\forall m,n\}$, the semi-infinite constraint Eq. (14) can be substituted equivalently by

$$\boldsymbol{\Phi}(x[n],y[n],s_m[n],\lambda_m[n])\geqslant 0,\ \forall n,\ m \tag{19}$$

$$\lambda_m[n]\geqslant 0,\ \forall n,\ m \tag{20}$$

3.3. Reformulation of non-convex constraints

Based on the previous two subsections, the problem Eq. (P3) can be lower bounded by the following form:

$$\text{(P4)}\quad \max_{\{\boldsymbol{q},\boldsymbol{u},\boldsymbol{a},\boldsymbol{I},\boldsymbol{L},\boldsymbol{s},\boldsymbol{\lambda}\}}\ \frac{1}{N}\sum_{n=1}^{N}p_0\beta_0\tilde{\gamma}_{\mathrm{fea}}[n] \tag{21a}$$

$$\text{s.t.}\quad \delta\sum_{n=1}^{N}\left[c_1\|u[n]\|^3+\frac{c_2}{\|u[n]\|}\left(1+\frac{\|a[n]\|^2}{g^2}\right)\right]+\Delta_k \leqslant\Gamma \tag{21b}$$

$$\|\boldsymbol{u}[n]\|\geqslant u_{\min},\ \forall n \tag{21c}$$

$$\boldsymbol{\Phi}(x[n],y[n],s_m[n],\lambda_m[n])\geqslant 0,\ \forall n,\ m \tag{21d}$$

$$\text{Eqs.(8c)-(8h), (10b), (13), (20)} \tag{21e}$$

However, the problem Eq. (P4) is still not a convex optimization due to the non-convex constraints Eq. (21b), Eq. (21c) and Eq. (21d). To this end, we propose an iterative algorithm to obtain an approximate solution of problem Eq. (P4) by leveraging the slack variables and SCA method.

By introducing slack variable $\boldsymbol{\tau}=\{\tau[n],\forall n\}$, we can rewrite constraints Eq. (21b) and Eq. (21c) as

$$\delta\sum_{n=1}^{N}\left[c_1\|\boldsymbol{u}[n]\|^3+\frac{c_2}{\tau[n]}\left(1+\frac{\|\boldsymbol{a}[n]\|^2}{g^2}\right)\right]+\Delta_k\leqslant\Gamma \tag{22}$$

$$\|\boldsymbol{u}[n]\|^2\geqslant u_{\min}^2,\ \forall n \tag{23}$$

$$\tau^2[n]\leqslant\|\boldsymbol{u}[n]\|^2,\ \forall n \tag{24}$$

$$\tau[n]\geqslant u_{\min},\ \forall n \tag{25}$$

The reformulation can also be proved by the similar argument to Lemma 1. With such transformation, the constraint Eq.(22) is now jointly convex with respect to $\{\boldsymbol{u}[n],\boldsymbol{a}[n],\tau[n]\}$. To solve the non-convex constraints Eq. (23) and Eq.(24), a local convex approximation is adopted. With the first-order Taylor expansion of $\|\boldsymbol{u}[n]\|^2$ at a feasible point $\boldsymbol{u}_{\mathrm{fea}}\triangleq[\boldsymbol{u}_{\mathrm{fea}}[1],\boldsymbol{u}_{\mathrm{fea}}[2]\ldots,\boldsymbol{u}_{\mathrm{fea}}[N]]$, we can reach the lower bound on $\|\boldsymbol{u}[n]\|^2$ as

$$\tilde{\boldsymbol{u}}[n]=\|\boldsymbol{u}_{\mathrm{fea}}[n]\|^2+2(\boldsymbol{u}_{\mathrm{fea}}[n])^{\mathrm{T}}(\mathbf{u}[\mathrm{n}]-\mathbf{u}_{\mathrm{fea}}[\mathrm{n}]) \tag{26}$$

Therefore, we can approximately convert the constraints Eq.(23) and Eq.(24) into the convex constraints

$$\tilde{\boldsymbol{u}}[n]\geqslant u_{\min}^2,\ \forall n \tag{27}$$

$$\tau^2[n]\leqslant\tilde{\boldsymbol{u}}[n],\ \forall n \tag{28}$$

To proceed, we are devoted to solving the non-convex constraint Eq.(21d). To tackle the non-linear terms of $x^2[n]$ and $y^2[n]$ in $c_m[n]$, we assume a feasible point $\boldsymbol{x}_{\mathrm{fea}}\triangleq[x_{\mathrm{fea}}[1],x_{\mathrm{fea}}[2]\ldots,x_{\mathrm{fea}}[N]]$ and $\boldsymbol{y}_{\mathrm{fea}}\triangleq[y_{\mathrm{fea}}[1],y_{\mathrm{fea}}[2]\ldots,y_{\mathrm{fea}}[N]]$. By using the following first-order Taylor expansions, the lower bound of $c_m[n]$ in Eq.(21d) can be obtained as

$$\tilde{c}_m[n]=-x_{\mathrm{fea}}^2[n]+2x_{\mathrm{fea}}[n]x[n]-2\tilde{x}_{\mathrm{j},m}x[n]+\tilde{x}_{\mathrm{j},m}^2-y_{\mathrm{fea}}^2[n]+2y_{\mathrm{fea}}[n]y[n]-2\tilde{y}_{\mathrm{j},m}y[n]+\tilde{y}_{\mathrm{j},m}^2+H^2-s_m[n] \tag{29}$$

3.4. Overall description and computation complexity

By summarizing the above developments, problem Eq. (P4) is approximately recast as

$$\text{(P5)}\quad \max_{\{\boldsymbol{q},\boldsymbol{u},\boldsymbol{a},\boldsymbol{I},\boldsymbol{L},\boldsymbol{s},\boldsymbol{\lambda},\boldsymbol{\tau}\}}\ \frac{1}{N}\sum_{n=1}^{N}p_0\beta_0\tilde{\gamma}_{\mathrm{fea}}[n] \tag{30a}$$

$$\text{s.t.}\quad \tilde{\boldsymbol{\Phi}}(x[n],y[n],s_m[n],\lambda_m[n])\geqslant 0,\ \forall n,\ m \tag{30b}$$

$$\text{Eqs. (8c)-(8h), (10b), (13), (20), (22), (25), (27), (28)} \tag{30c}$$

with

$$\tilde{\boldsymbol{\Phi}}(x[n],y[n],s_m[n],\lambda_m[n])=\begin{bmatrix}\lambda_m[n]+1 & 0 & \tilde{x}_{\mathrm{j},m}-x[n]\\ 0 & \lambda_m[n]+1 & \tilde{y}_{\mathrm{j},m}-y[n]\\ \tilde{x}_{\mathrm{j},m}-x[n] & \tilde{y}_{\mathrm{j},m}-y[n] & -\varepsilon_m^2\lambda_m[n]+\tilde{c}_m[n]\end{bmatrix}$$

It is not difficult to see that the objective function is concave, and all the constrains are convex. Therefore, Eq. (P5) is a convex optimization problem, which can be optimally solved in polynomial time by the Interior-point Method (IPM).[33]

In conclusion, since Eq. (P1) is a non-convex semi-infinite optimization problem, it is exceedingly difficult to obtain the global optimal solution. Based on the simplifications and approximations in previous parts, we recast the formulation into a tractable form Eq. (P5), which is more amenable for optimization. It readily follows that the objective value of Eq. (P5) gives a lower bound to that of Eq. (P1), and they are equal only at the given point $(\boldsymbol{x}_{\text{fea}}, \boldsymbol{y}_{\text{fea}}, \boldsymbol{u}_{\text{fea}}, \boldsymbol{I}_{\text{fea}}, \boldsymbol{L}_{\text{fea}})$. As a result, an efficient optimal technique for problem Eq. (P1) is concluded in (Algorithm 1). It is worth pointing out that the objective value of problem Eq. (P5) is finite, and the optimal value obtained by each iterative is non-decreasing, which ensures the convergence of Algorithm 1.

Next, we are devoted to evaluating the computation complexity of the proposed algorithm. The complexity of Algorithm 1 consists of three parts, namely, the required iterations for SCA method, the complexity of iterative and the computation cost for per-iteration. We assume that the iteration number of the SCA algorithm is T_{ite} and the accuracy of IPM for each iteration is ℓ. It is not difficult to see that problem Eq. (P5) contains MN Linear Matrix Inequalities (LMIs) of size 3, $(M+4)N+1$ LMIs of size 1 and $5N$ Second-order Cone (SOC) constraints of dimension 2. Meanwhile, the numbers of decision variables are on the order of N. Based on the complexity analysis in,[34] the total computation complexity of Algorithm 1 is given by

$$\mathcal{O}(T_{\text{itc}}\sqrt{(4M+11)N+1}ln(\ell^{-1})N[28MN+10MN^2+5N^2 + 25N+1])$$

4. Numerical results

In this section, numerical simulations are provided to validate the effectiveness of our proposed trajectory optimization algorithm (denoted as the robust scheme) and draw some essential insights. For verifying the enhancement of the robust scheme, we consider following two benchmark methods. (A) non-robust scheme[34] (B) fixed trajectory scheme.[23] Specifically, the non-robust scheme can be obtained from Section III, where we treat the estimated information of each jammer as perfect information, i.e., $\Delta\boldsymbol{w}=\boldsymbol{0}$, $\Delta p_{\text{j},m}=0$, $\forall m$, and then calculate the worst-case average SINR obtained in the bounded error model. The fixed trajectory scheme makes the UAV fly straightly from the initial point to final point with the constant velocity. To demonstrate the efficacy of the proposed algorithm, we deliberately set the energy consumption constraint to be sufficient to ensure the UAV can complete the mission under each scheme.

In the simulation, we consider a UAV mission execution scenario where the UAV flies from the horizontal starting $_{point}$ $\boldsymbol{q}_{\text{I}}=[100,\ 100]$ m to the endpoint $\boldsymbol{q}_{\text{F}}=[800,\ 800]$ m at a specific altitude $H=100$ m and maintains communication with the GCU. Furthermore, it is assumed that the whole flight duration for completing the task is $T=30$ s and each time slot is $\delta=1$ s. Besides, we assume the location of GCU is $[0,\ 0,\ 0]$ m and the transmit power is set to $p_0=10$ W. There are $M=3$ jammers distributed across the ground and the estimated horizontal coordinates are $\tilde{\boldsymbol{w}}_{\text{j},1}=[700,\ 800]$ m, $\tilde{\boldsymbol{w}}_{\text{j},2}=[500,\ 600]$ m and $\tilde{\boldsymbol{w}}_{\text{j},3}=[300,\ 200]$ m, respectively. Meanwhile, the estimated transmit power of jammers are all set to $\tilde{p}_{\text{j},1}=\tilde{p}_{\text{j},2}=\tilde{p}_{\text{j},3}=0.1$ W. For simplicity, all jammers' estimation power error is set to $\xi_m=0.02$ W. If not otherwise specified, other corresponding simulation parameters are summarized in Table 1.

We first demonstrate the convergence performance of the proposed robust trajectory scheme, as well as the non-robust scheme, by setting all jammers' estimation location error with the bound of $\varepsilon_m=10$ m. As we can see in Fig. 2, the objective function value obtained by two algorithms increases rapidly with the number of iterations, and then converges within around 4 iterations for non-robust scheme and 5 iterations for robust scheme. This is due to the fact that the proposed robust trajectory scheme requires more computational costs to solve the semi-infinite constraint evoked by the uncertainty of jammer information, which makes it more difficult to converge.

Fig. 3 depicts the optimized UAV trajectory obtained by different schemes, under different setups of Γ with all jammers' estimation location error set as $\varepsilon_m=40$ m. It is observed that for the robust scheme and non-robust scheme, the UAV tries

Table 1 Simulation parameters.

Description	Value
UAV maximum speed $V_{\max}$ (m/s)	50
UAV minimum speed $V_{\min}$ (m/s)	3
UAV maximum acceleration $a_{\max}$(m/s^2)	5
Channel gain β_0 (dB)	−30
Noise Power σ^2 (dBm)	−80
Path-loss exponent α	2
Maximum tolerance ℓ	10^{-4}
Parameter for energy consumption c_1	9.26×10^{-4}
Parameter for energy consumption c_2	2250
Gravitational acceleration g (m/s^2)	9.8

Algorithm 1. SCA-based trajectory planning algorithm for problem (P1)

1. Construct the initial feasible solution $(\boldsymbol{x}^{(0)}, \boldsymbol{y}^{(0)}, \boldsymbol{u}^{(0)}, \boldsymbol{I}^{(0)}, \boldsymbol{L}^{(0)})$. Let $k=0$.
2. **Repeat**
3. Set $k \leftarrow k+1$.
4. Set the feasible points $\boldsymbol{x}_{\text{fea}}=\boldsymbol{x}^{(k-1)}$, $\boldsymbol{y}_{\text{fea}}=\boldsymbol{y}^{(k-1)}$, $\boldsymbol{u}_{\text{fea}}=\boldsymbol{u}^{(k-1)}$, $\boldsymbol{I}_{\text{fea}}=\boldsymbol{I}^{(k-1)}$, $\boldsymbol{L}_{\text{fea}}=\boldsymbol{L}^{(k-1)}$, then solve the problem (P5) and denote the optimal solution as $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{u}^*, \boldsymbol{\Gamma}^*, \boldsymbol{L}^*)$.
5. Update the variable $\boldsymbol{x}^{(k)}=\boldsymbol{x}^*$, $\boldsymbol{y}^{(k)}=\boldsymbol{y}^*$, $\boldsymbol{u}^{(k)}=\boldsymbol{u}^*$, $\boldsymbol{I}^{(k)}=\boldsymbol{\Gamma}^*$, $\boldsymbol{L}^{(k)}=\boldsymbol{L}^*$.
6. **Until** the fractional increase of the objective value is below a given threshold ℓ.

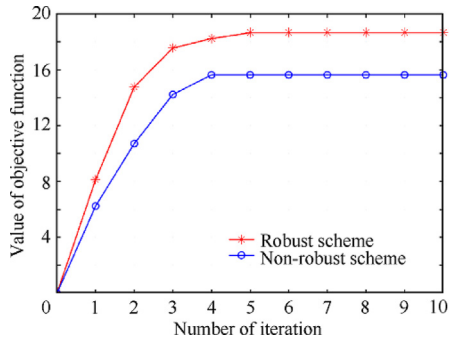

Fig. 2 Convergence performance of the proposed algorithm ($\varepsilon_m = 10$ m).

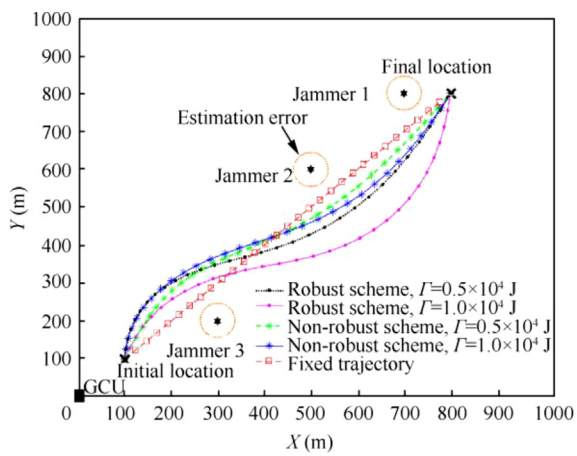

Fig. 3 UAV's trajectory for $\varepsilon_m = 40$ m under different values of Γ.

to fly away as far as possible from the jammers. The reason is that when the distance from the jammers increases, the intensity of jamming signals accordingly weakens. Thus, moving further away from the jammers is conducive to enhancing the quality of communication. Furthermore, when the flight energy consumption budget becomes larger (e.g., $\Gamma = 1.0 \times 10^4$ J (Joule)), it is interesting to note that the UAV notably adjusts the trajectory further to the jammers relative to the case with $\Gamma = 0.5 \times 10^4$ J. This observation suggests that more energy consumption allows the UAV to find better trajectory to combat interference and improve the communication performance. Besides, we notice that the proposed robust scheme obtains the better flight trajectory, as compared to that of non-robust scheme. The reason for this outcome is that the proposed robust trajectory scheme fully exploits the uncertainty of the jammers to effectively support reliable communication, while the other schemes do not take these features into consideration.

Fig. 4 compares the UAV trajectories by applying different schemes with respect to estimation location error ε_2 when the energy consumption budget $\Gamma = 1.0 \times 10^4$ J. As expected, the proposed robust scheme always outperforms all the other benchmark methods in the entire flight duration, which shows the advantage of the robust trajectory optimization in system performance gain. Moreover, we note that the trajectory of non-robust scheme remains unchanged regardless of the estimation location error. It is also observed that the distance to jammer 2 of the robust scheme notably increases with ε_2. As a result, it appears that for UAV network with a large uncertainty in location estimation, our proposed robust scheme is particular appealing from a trajectory optimization perspective.

To further demonstrate the UAV's behavior, Figs. 5 and 6 show the corresponding time-varying speed of the trajectory in Figs. 3 and 4, respectively. One can see that as the available energy consumption or the uncertainty of location estimation increases, the gap between different schemes become larger, which validates the effectiveness of the proposed robust scheme in manipulating highly uncertain scenarios. Moreover, for the proposed robust scheme and non-robust scheme, it is observed that UAV starts at a relatively low speed for the sake of maintaining high-quality communication with close GCU. Furthermore, UAV gradually moves away from the GCU and approaches the jammers, thereby resulting in a less favorable propagation condition. At this time, the intensity

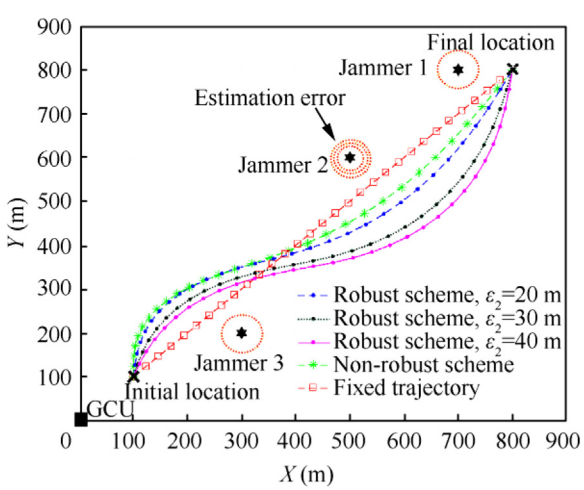

Fig. 4 UAV's trajectory for $\Gamma = 1.0 \times 10^4$ J under different values of ε_2.

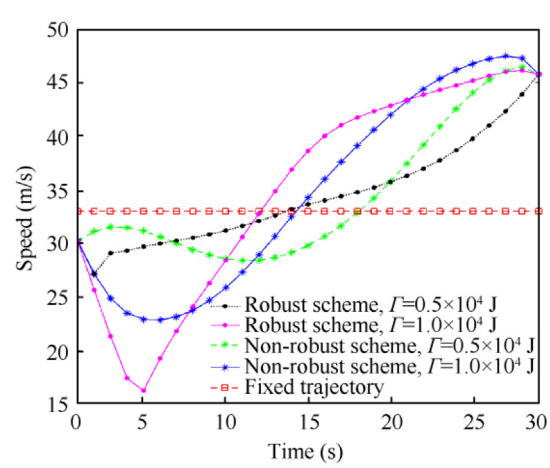

Fig. 5 UAV's flight speed for $\varepsilon_m = 40$ m under different values of Γ.

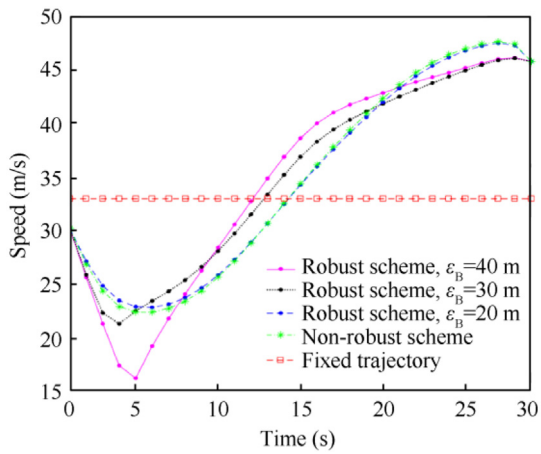

Fig. 6 UAV's flight speed for $\Gamma = 1.0 \times 10^4$J under different values of ε_2.

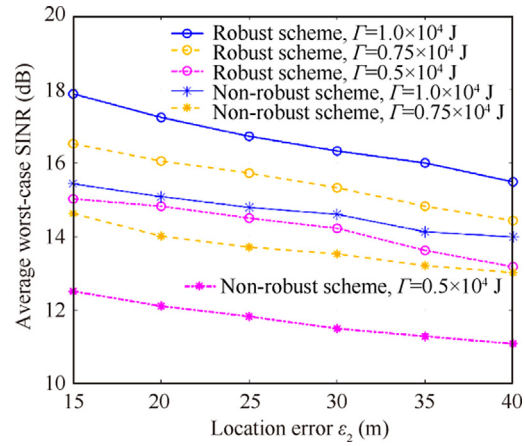

Fig. 8 Average worst-case SINR versus the jammer 2's estimation location error ε_2.

of jamming plays a dominant role in the communication performance. Intuitively, it is found that when the UAV approaches the jammers, it adjusts its flight direction to avoid the jamming (shown in Figs. 3 and 4), while increases the speed to fly away from the jammers, which is consistent with the practical flight scenarios of UAV.

In what follows, we are devoted to comparing the performance between the proposed robust scheme and non-robust scheme. Fig. 7 shows the UAV's average worst-case SINR achieved by two algorithms versus the energy consumption threshold. One can see that the robust algorithm always achieves higher average worst-case SINR, by up to about 115%, compared to the non-robust algorithm. This performance gain is due to the fact that the robust scheme makes the flight trajectory more reasonable, which can reduce the impact of external jamming and enhance the average worst-case SINR. What's more, we notice that the average worst-case SINR of two methods first increase with Γ when Γ is small and then remain invariant when Γ exceeds a certain level. The reasons are twofold. For one thing, when the energy consumption budget is small, the increase in Γ allows the UAV to explore more appropriate trajectories, leading to an increase in the SINR. For another, when Γ is large enough, the optimal flight trajectory could be obtained at a certain energy consumption below the threshold, thus further increasing of Γ could not enlarge the communication quality any more.

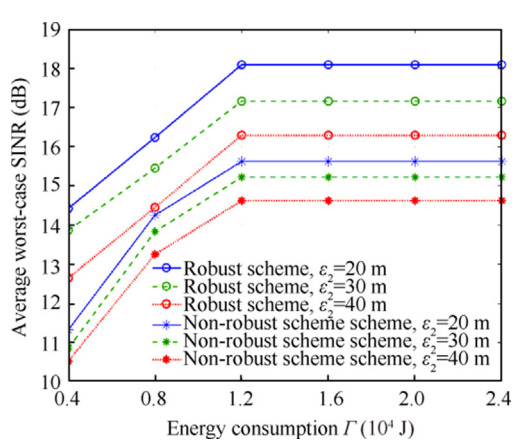

Fig. 7 Average worst-case SINR versus the energy consumption threshold Γ.

Finally, we consider the effect of the jammers' location uncertainties on the communication performance. Fig. 8 shows the UAV's average worst-case SINR of the robust scheme and non-robust scheme versus the radius of the uncertain region of jammer 2 ε_2 under different energy consumption budget. It is observed that the SINR of the two algorithms decrease as ε_2 increases. As can be seen in Fig. 4, although the trajectory of non-robust scheme remains unchanged under any radius of uncertain region, the average worst-case SINR will decrease with the increase of the location error. Since the non-robust scheme ignores the uncertainty of the error, the robust algorithm offers superior performance over the non-robust algorithm. There results validate the efficiency of the proposed robust scheme in handling different scenarios for uncertainty, thereby endowing added flexibility to the design of the UAV system.

5. Conclusions

In this paper, we have studied the reliable communication in a GCU-to-UAV communication network, where a UAV is operated by a GCU to perform certain tasks in the presence of multiple jammers with imperfect power and location information. We aim at maximizing the UAV's average worst-case SINR by optimizing the trajectory over a finite flight duration, subject to the mobility and energy consumption constraints. To tackle the formulated non-convex semi-infinite problem, we develop an iterative algorithm to obtain the solution with the aid of slack variables, S-procedure and SCA method. Simulation results show that the proposed robust trajectory scheme can significantly enhance the communication performance as compared to other benchmark schemes. This paper aimed to shed more light on the trajectory design of the UAV communication system, which can be extended in several interesting directions for the future work, including on-line trajectory optimization, intelligent resource coordination, as well as advanced priority-aware design, through the application of Artificial Intelligence (AI), e.g., reinforcement learning, "learning to optimize".

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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